HO ZHU YI

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Academic Qualifications

London, UK — **Imperial College London** — **Oct 2015 – Jun 2019**

- **Major:** MEng in Civil Engineering (First Class Honours 87%)
- **Awards:** Dean's list recipient (top 10% in years 1-4), ICE Prize (best in third and fourth years), Letitia Chitty Prize (greatest merit in the third year), Peter Fraenkel Prize (best third year Group Design Project)
- **Programming Courses:** Computational Methods 1 & 2, Operational Research and Systems Analysis

Professional Experience

Full Stack Developer — **HSP Power Systems Sdn. Bhd.** — **Feb 2026 – Present**

Full-stack development of the ERP used by 100+ staff of a generator set supplier (ASP.NET Core, Angular, SQL Server, AWS).

- Led the first build of the order-to-delivery flow, from sales quotations and purchase orders to goods receipt, assembly and delivery, end to end from SQL Server schema and ASP.NET Core API to Angular UI, working directly with users.
- Designed the front-end core every module is built on: a central edit store behind all editing and saving, and config-driven forms, data tables and detail pages, so new modules are mostly configuration; introduced TanStack Query for server state.
- Built Claude skills that trace data issues through legacy FoxPro code, stored procedures and scanner backends, then write SQL fixes in three steps (diagnose, fix, verify), freeing half a graduate's time.
- Cut monthly sales reporting from a week of manual work to minutes: built a prospect tracker (six milestones to order) in the ERP and a Python pipeline (marimo, Parquet) feeding Power BI dashboards of sales targets, prospects and receivables.
- Designed role-based access control: Entra ID roles map to fine-grained permissions in one central table read by both the API and the Angular UI, so code checks permissions, never roles, and access is managed per role, not per user.
- Built Excel upload for 200,000-row vendor price lists: Web Worker parsing, batch validation and batch database writes.
- Built a reusable email notification service with queued daily digests; introduced FluentValidation for API validation.

Computational Structural Engineer — **Arup Jururunding Sdn. Bhd. / Arup Singapore Pte. Ltd.** — **Aug 2019 – Jan 2026**

Structural engineer on high-rise towers in Malaysia and Singapore, including Merdeka 118 (world's second tallest).

- Built Python wrappers over the .NET and COM APIs of commercial analysis programs, linked by one Polars data schema.
- Automated reinforcement design of columns, walls, coupling beams and piles for whole towers using numerical optimisation.
- Made ray–triangle intersection checks 100× faster than the commercial tool with vectorised NumPy and Numba.
- Built a Jupyter logging framework with automated Altair charts to track and compare models across dozens of revisions.
- Led engineers across the Shanghai, Kuala Lumpur and Singapore offices to a successful tender, coordinating consultants.

Relevant Projects

- **Bizzy (2026)**: business app for a physiotherapy practice, in production: bookings synced with Google Calendar, packages, invoicing and WhatsApp payment requests. React PWA, TypeScript, Go, AWS Lambda, DynamoDB, CDK, GitHub Actions. [Link]
  - Property-based state-machine tests check every business rule, including against production data before each deploy.
  - Transactional, rule-checked writes keep data valid; invoices are reproducible, hashed PDFs locked in S3 for 7 years.
- **Owlmanac (2025–):** blog of interactive posts computed in the browser. Astro, Vue, Rust/WASM, DuckDB, marimo. [Link]
  - **Mapping the Milky Way:** analysed Gaia DR3 star data to map the galaxy's structure (Polars, Astropy, Altair).
  - **Solving puzzles with SAT:** a Rust CDCL solver behind an in-browser puzzle game with hints, and 6,000 generated puzzles.
- **Bingeon (2025):** movie discovery with Gemini picks, inside AWS and Google free tiers. React, Go, Lambda, DynamoDB. [Link]
- **Structural design pipeline (2020–2026):** Python library automating Eurocode 2 concrete design across ETABS, SAFE and AdSec, with clause-level tests and revision comparison; used on high-rise towers in Malaysia, plus smaller tools. [Link]
- **Filtration network model (2019, 86%):** Imperial final-year project. An object-oriented MATLAB model of soil particles moving through dam filters, with a sparse-matrix Stokes flow solver, testing filter design rules over 504 combinations. [Link]

Professional Certifications

Full Stack Development with AI (99.8%) — **National University of Singapore** — **Apr – Oct 2025**

- Full-stack web development (React, TypeScript, Express, Flask, SQL and NoSQL) and machine learning (scikit-learn, TensorFlow).

CS50 Intro to Computer Science — **Harvard University** — **Apr – Oct 2025**

- Programming, data structures and algorithms with C; frontend dev with HTML, CSS, JS; backend dev with Python (Flask), SQL.

Skills

Languages: TypeScript, C#, Go, Python, Rust, SQL, JavaScript, HTML, CSS, C, VBA, MATLAB.
Frontend: Angular, React, Vue, Astro, TanStack Query, Angular Material, Tailwind, shadcn/ui, Web Workers, WebAssembly, Vitest.
Backend and data: ASP.NET Core, EF Core, FluentValidation, SQL Server, DynamoDB, SignalR, REST; Polars, marimo, Power BI.
Cloud and delivery: AWS (CDK, Lambda, App Runner, ECS Fargate, S3, CloudFront, Cognito), Docker, GitHub Actions, Claude Code.
Spoken: English, Mandarin, Malay (fluent).